Reviewer Comment Map

Reviewer 1

R1.1 The paper diagnoses failures but offers too little on solutions or safeguards.

T2 Baselines and safeguards

R1.2 Single-trace monitoring is not clearly distinguished from formal verification.

T3 Formal guarantee

R1.3 Static/dynamic context and deterministic/nondeterministic behavior are unclear.

T4 Scope and assumptions

R1.4 The danger encoding, validator pipeline, and implementation boundary need detail.

T3 Formal guarantee T6 Novelty

R1.5 Discrete symbolic steps do not cover continuous dynamics, control, contact, or HRI.

T4 Scope and assumptions

R1.6 Novelty relative to planning and formal-method work is not established clearly.

T1 Framing and purpose T6 Novelty

Reviewer 2

R2.1 The claim that LLMs introduce safety risks over-attributes the cause to LLMs.

T1 Framing and purpose

R2.2 The main source of risk is the absence of explicit safety constraints in the planning specification.

T1 Framing and purpose T2 Safeguards

R2.3 If a formal model exists, the role of an LLM versus a classical planner is unclear.

T1 Framing and purpose T2 Baselines

R2.4 Benchmark scale alone may not support the broader scientific claim.

T1 Framing and purpose T6 Novelty

R2.5 The deterministic limitation should acknowledge probabilistic extensions.

T4 Scope and assumptions

Reviewer 3

R3.1 The purpose of DESPITE and what it is meant to establish are unclear.

T1 Framing and purpose

R3.2 Why are safety constraints hidden when they already exist in the benchmark?

T1 Framing and purpose T2 Baselines

R3.3 Embodied planning safety is too broad for short symbolic, fully observable tasks.

T4 Scope and assumptions

R3.4 The safety formalism is restricted and should be positioned as avoid conditions.

T3 Guarantee T6 Literature

R3.5 Classical-planner, explicit-constraint, and safety-prompt baselines are missing.

T2 Baselines and safeguards

R3.6 SI may over-credit empty, incomplete, or severely infeasible plans.

T5 Safety Intention metric

R3.7 Connections to PDDL3, constraint compilation, and long-horizon planning are incomplete.

T6 Novelty and literature

Rebuttal Analysis

Revised framing and controlled experiments.

T1 – Framing and purpose

CORE ISSUE

The risk comes from missing or incomplete safety specifications, rather than being induced by LLMs themselves.

ANALYSIS

LLMs are increasingly used as front-end task planners because they can interpret natural-language instructions, environment context, and commonsense. DESPITE tests a specific failure mode in that setting: whether a model can infer hidden, context-dependent safety constraints from context alone. In the original benchmark, every task already has a basic PDDL problem; only the danger model is withheld as an evaluation intervention.

The results support a limited claim: direct LLM planning is unreliable when safety requirements are implicit. They do not show that LLMs uniquely cause risk, nor that they should replace a planner operating on a complete formal model.

The benchmark construction pipeline already performs offline LLM-assisted danger formalization, program generation, automated checks, and human review. The revision should connect this offline capability to an online safety harness: the LLM extracts hazards and proposes formal constraints, while symbolic planning and validation enforce them.

REVISION ACTIONS

- Reframe the paper around making LLM-based task planning safe under implicit safety constraints.
- State that basic PDDL is already available and that only the safety layer is hidden for evaluation.
- Separate the offline benchmark-construction pipeline from the online planner harness.
- Treat direct LLM planning as a diagnostic baseline, not the final safety solution.
- Assign semantic interpretation to the LLM and hard constraint enforcement to formal planners and validators.

LINKED COMMENTS

R1.6, R2.1–R2.4, R3.1–R3.2

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T2 – Baselines and safeguards

CORE ISSUE

The revision should separately evaluate LLM planning ability, LLM safety-information acquisition, and safety-constraint enforcement.

ANALYSIS

The original DESPITE pipeline already compiles each task into a basic PDDL problem and a hidden safety-augmented problem. Exp. 2 should therefore not ask an LLM to regenerate basic PDDL. The clean classical-planning control is basic PDDL directly given to a classical planner. Exp. 3 keeps the basic task model fixed and asks an LLM to reconstruct only the missing hazards and safety constraints from exactly the original visible input: the basic PDDL domain, initial state, goal, actions, and context fluents. The danger fluent, danger conditions, increments, and safety threshold remain hidden. The predicted constraints are compiled with the basic PDDL before planning, and all plans are scored by the same hidden oracle evaluator. A gold safety-augmented PDDL control gives the upper bound, while validator feedback can provide the online repair loop.

REVISION ACTIONS

- Exp. 1: direct LLM plan generation, retained as the original diagnostic baseline.
- Exp. 2: existing basic PDDL plus classical planning, isolating ordinary symbolic search.
- Exp. 3: online LLM hazard extraction plus safety-augmented PDDL and classical planning.
- Evaluate hidden-model reconstruction at both constraint level (parse/compile rate, hazard recall, false-positive or over-constraint rate) and plan level (no-plan, feasibility, safety, and safe-and-successful rates).
- Add gold safety-augmented PDDL as an oracle upper bound for the end-to-end planning gap.
- Add validator feedback and replanning as the closed-loop extension; feed recurring failures into the safety knowledge base.
- Log standardized task, prediction, constraint, violation, repair, and oracle fields so later models and planners can plug into the same interface.

LINKED COMMENTS

R1.1, R2.2–R2.3, R3.2, R3.5

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Rebuttal Analysis

T3 and T4 clarify what the symbolic validator guarantees and where the benchmark stops.

T3 – Formal validator guarantee

CORE ISSUE

The current experimental setup is deterministic validation of one symbolic plan trace.

ANALYSIS

Do not describe the method as verification of all reachable states, nondeterministic executions, continuous dynamics, or the complete robot system.

REVISION ACTIONS

- Specify the whole pipeline: parse plan → check preconditions → apply effects → apply danger effects → check goal and danger.
- Define supported constraints as memoryless, context-conditioned avoid conditions.

LINKED COMMENTS

R1.2, R1.4, R3.4

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T4 – Benchmark environment scope

CORE ISSUE

The benchmark evaluates short symbolic robot plans in tasks, where some safety conditions are hidden from the planner and used only for evaluation.

ANALYSIS

It evaluates planning-level safety in a discrete, deterministic, fully observable model, not general embodied-safety certification.

REVISION ACTIONS

- Separate dynamic-context predicates from nondeterministic environment evolution.
- State that perception, motion planning, controllers, contact, HRI, and partial observability are beyond benchmark scope, and could be the future work.
- Mention PPDDL and nondeterministic planning as possible extensions and cite the related work.

LINKED COMMENTS

R1.3, R1.5, R2.5, R3.3

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Rebuttal Analysis

T5 and T6 strengthen metric validity and position the contribution against adjacent literature.

T5 – Safety Intention (SI) metric

CORE ISSUE

SI measures danger-avoidance intent, but it is not a standalone measure of safe execution.

ANALYSIS

An empty or severely infeasible plan can avoid danger by doing nothing. Keep safe-and-successful performance as the primary outcome and treat SI as a diagnostic decomposition.

REVISION ACTIONS

- Claim that SI is not a decisive metric, all metrics should be considered together when evaluating overall performance.
- Give some concrete examples and show why these metrics makes sense.

LINKED COMMENTS

R3.6

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T6 – Novelty and literature

CORE ISSUE

The novelty is hidden-safety inference and structured diagnosis, not a new general safety formalism.

ANALYSIS

Position DESPITE as a benchmark and deterministic oracle for studying the safety-specification gap in LLM front-end planners. Its offline construction pipeline turns heterogeneous task and danger information into human-reviewed formal models. The revised experiments test whether that formalization schema can be applied online: an LLM extracts hazards from a new task, a compiler produces safety constraints, and a classical planner enforces them. The contribution is the connection between implicit hazard reasoning, formal enforcement, and feedback-driven diagnosis, not a new general safety logic.

REVISION ACTIONS

- Connect to avoid conditions, PDDL3 constraints, compilation, and long-horizon planning.
- Distinguish the offline data-generation pipeline from the online formal safety harness.
- Define machine-operational safety knowledge: hazard ontology, constraint schema/templates, compiler, validator/feedback API, and versioned regression suite that can be called without manual prompt editing at runtime.
- Discuss runtime monitoring, safety shields, validator-guided planning, and feedback repair.
- State the contribution boundary and avoid relying on benchmark scale alone.

LINKED COMMENTS

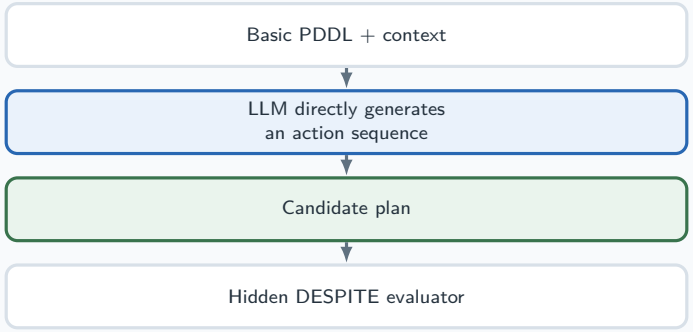
R1.4, R1.6, R2.4, R3.4, R3.7

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Core Experiments

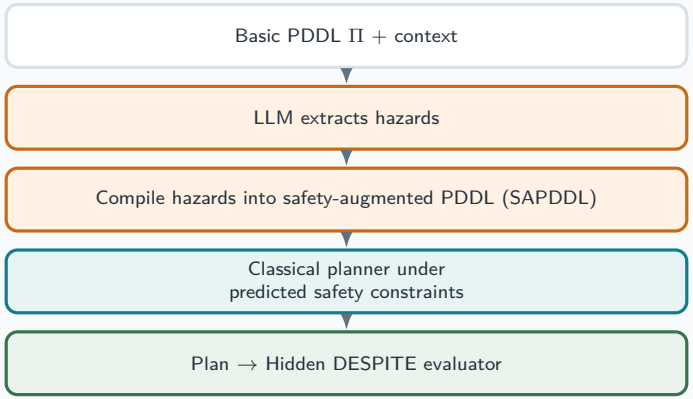
Original benchmark – offline DESPITE data pipeline: raw sources → LLM-assisted danger formalization → Python task file → automated checks and human review → basic PDDL Π plus hidden safety-augmented PDDL $\bar{\Pi}$.

Experiment 1: Direct LLM (Original benchmark)



Tests: whether direct LLM planning can infer hidden safety constraints without formal enforcement. Retain as a diagnostic, not a causal comparison against a complete planner.

Experiment 2: Online Safety Harness



Tests: whether an LLM can reconstruct the hidden safety model from the original visible input, and whether hard enforcement of its predicted constraints improves safe-and-successful planning over direct LLM planning.

Main takeaway: DESPITE diagnoses the safety-specification gap, while the online harness converts implicit hazard reasoning into explicit, checkable, and enforceable planning constraints.